When Tooth-Numbering Detectors Fail Together: A Position-Only Baseline and Closed Gaps on Panoramic X-rays

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Abstract. Tooth-numbering models for panoramic X-rays report high accuracy, but the X-ray format places each tooth in a predictable spot, so part of that accuracy could come from position rather than appearance. We measure this with a position-only baseline: a classifier that sees each tooth's bounding box and no pixels. On 425 uncropped panoramic X-rays (11,602 teeth) with five-fold cross-validation grouped by X-ray, it numbers 78.5% of teeth correctly. Three detectors of different design (YOLOv8x, RT-DETR-l, Faster R-CNN) beat it by 14.3 to 16.7 percentage points, on all 32 tooth numbers, and the gap replicates on two further splits. Their errors still follow position. On 1.4% of teeth all three detectors give the same wrong number, and in 91% of these the position-only model gives it too, against under 7% by chance. They sit next to a missing tooth five times as often as correct teeth and usually take the missing tooth's number, and they occur where the neighbors have closed the space (AUROC 0.78). Closing a gap on purpose reproduces them: when a neighbor is moved into an erased tooth's space, all three detectors give it the missing number 21% of the time, against 0.2% left open. On 634 DENTEX X-rays from other clinics, detectors trained there beat position alone by 22 to 25 points; their shared errors give the position-only answer 85.5% of the time, link weakly to missing teeth, and closing a gap reproduces them at 12%. We release the per-tooth table and a scorer.

Keywords: Tooth numbering · Panoramic radiography · Shortcut learning · Error consistency

1 Introduction

Assigning each tooth its FDI number is the first step of most automated dental charting: a caries or a periapical lesion found by a detector is only useful once it is attached to the right tooth. Public benchmarks such as DENTEX [8] score numbering together with detection, and reported accuracies are high.

Panoramic X-rays are a strongly standardized format. The patient bites on a guide, the arch runs left to right across the image, and each tooth number has a typical location. This invites a shortcut [6]: a model can get many numbers right from where a box sits, without reading the tooth. Such a shortcut would

be invisible in aggregate accuracy and would fail on the patients who differ from the typical arch, for example after extractions.

We ask how much of tooth numbering position alone explains, and whether detectors fail where position fails. Our contributions: (1) a position-only baseline, trained in seconds on six box features, that numbers 78.5% of teeth correctly under grouped cross-validation, with three detectors beating it by 14 to 17 points on every tooth number; (2) an error analysis showing that the detectors' shared errors are position-like; (3) interventions and checks (image shift, context masking, tooth erasure, a space-closure measure and gap closure on purpose) that place these errors at closed gaps, next to a missing tooth whose neighbors have moved together so the arch looks complete; and (4) a released per-tooth benchmark with folds, reference predictions and a scorer.

2 Related Work

Tooth numbering. Detection-based numbering on panoramic X-rays is standard, with DENTEX [8] and hierarchical diffusion detectors [9] as recent examples and many DENTEX challenge entries building on YOLO or DETR variants [10,15,4]. Some methods use the arch layout as an explicit prior [1]. Evaluations report accuracy or mAP; we found none that compares a model with what position alone achieves on the same teeth.

Shortcut learning. Models can rely on cues that correlate with the label in training data, from skin markings in dermoscopy [19] to site-specific features in chest X-rays [5] and segmentation [14]; see also [11,3]. These studies show a shortcut by removing or altering the cue. We use the same logic for a cue that is partly legitimate: position does carry information about tooth number.

Error consistency. Whether two models fail on the same inputs beyond chance is measured by error-consistency kappa [7]. We go one step further and compare the wrong answers themselves.

3 Methods

Data and splits. UFBA-425 [2] contains 425 panoramic X-rays (512×512 pixels) with per-tooth masks labeled by FDI number (32 classes, third molars included). Each tooth's box is the full extent of its mask, giving 11,602 teeth. We use the original X-rays only; an augmented export of this data adds randomly cropped copies, which move teeth in the frame and lowered position-only accuracy by about 8 points in our earlier splits. X-rays are split into five folds stratified by the dataset's categories; each training fold holds out 15% of its X-rays for model selection, and each X-ray is tested once by models that never saw it. Two further splits (other seeds) share at most about a quarter of any test fold with the first.

Models. The position-only baseline is the gradient-boosted tree classifier of scikit-learn (HistGradientBoosting, default settings) on six features of the labeled box: center x and y, width, height, area and aspect ratio, relative to

image size. It never sees pixels. The detectors are YOLOv8x [12], RT-DETR-l [20] and Faster R-CNN [17] with a ResNet-50 FPN backbone [13], all from COCO weights: a single-stage CNN, a transformer and a two-stage CNN, from two separate code bases. Each is trained for 30 epochs at 640 pixels with horizontal flips off (a flip swaps left and right tooth numbers); Faster R-CNN uses SGD with a learning rate scaled to its batch of 2. The epoch with the best validation fitness ($0.1\,\mathrm{mAP}_{50} + 0.9\,\mathrm{mAP}_{50:95}$) is kept.

Scoring. Detections with confidence at least 0.5 are matched one-to-one to labeled teeth at IoU at least 0.5, highest confidence first, regardless of class. A tooth with no match counts as wrong. Top-1 is the share of labeled teeth matched and given the right number, which puts the detectors and the baseline (which always answers) on the same teeth. We also report numbered F1, which counts extra boxes. AP50 and AP50:95 are in the supplement; top-1 stays the headline because the position-only baseline has no scores. CIs are 95% intervals from 10,000 bootstrap draws of X-rays, paired for differences. Each experiment's decision rule was written and frozen before it was run. A tooth is a *joint failure* when all three detectors and the position-only model are wrong and the detectors give the same answer. The *position match* is the share of joint failures where the position-only model also gives that answer; its chance level comes from shuffling the position-only answers among the joint failures.

Interventions. *Shift:* the whole image is translated by $\pm 10\%$ of its width (about two tooth widths), labels moved to match. *Context masking:* everything outside a tooth's box widened by k box widths and heights ($k = 0.5, 1, 2$) is blacked out. *Missing neighbor:* an adjacent arch position has no labeled tooth. *Tooth erasure:* for up to three teeth per X-ray with both neighbors present, the tooth's mask (dilated 3 px) is inpainted [18], and we ask whether a correctly numbered neighbor now takes the erased tooth's number; controls are the teeth two places away and a sham inpaint over bone. *Space closure:* for each missing tooth with both neighbors present, S is the shortest distance between the neighbors' masks over the missing tooth's median width (0 when they touch). *Gap closure:* after erasing a target tooth, its distal neighbor is cut out and pasted shifted toward the gap by half or all of the distance at which its mask touches the tooth on the other side, with its box moved to match.

4 Results

Position explains most; detectors add 14 to 17 points. Table 1. Position alone numbers more than three in four teeth. Each detector beats it on all 32 tooth numbers; the per-number gap is smallest at the third molars (4.3 to 5.8 points) and largest at the lower central incisors (up to 25.5). Given the detectors' own predicted boxes instead of labeled ones, the baseline does slightly better (79.0 to 80.2%) and the gap is 14.5 to 16.2 points, so labeled boxes do not flatter it. On two further splits every gap falls inside the first split's CI. YOLOv8x and RT-DETR-l tie on top-1 (difference 0.17 against a minimum detectable difference

Table 1. Top-1 on 11,602 teeth (split 0, 95% CI) and the gap over the position-only model. F1 counts extra boxes. Splits 2, 3: gap on two further random splits.

	Top-1 (%)	Gap (points)	Numbered F1	Splits 2, 3 gap
Position-only	78.5 (76.9–80.0)			
YOLOv8x	95.0 (94.2–95.8)	16.5 (15.1–17.9)	95.0	15.8, 16.4
RT-DETR-l	95.2 (94.4–95.9)	16.7 (15.3–18.1)	92.3	15.9, 16.4
Faster R-CNN	92.8 (91.9–93.7)	14.3 (13.0–15.7)	89.6	13.8, 14.3

of 0.50), but RT-DETR-l leaves about 1.9 unmatched boxes per X-ray against 0.4, which lowers its F1 by 2.7 points.

Shared errors follow position. 161 teeth (1.4%) are joint failures. In 91.3% of them the position-only model gives the same wrong number, against a permutation 95th percentile of 6.8%, and 94.4% of the shared wrong numbers are one place away in the same quadrant. The detectors are about as confident on these teeth as on correct ones (AUROC 0.53 to 0.66), so a confidence threshold does not find them, and neither does a rule based on the detectors' second choices. About 10% of joint failures may be label errors by confident learning [16]; without them the position match is 91.7%.

Not absolute position, not distant context. Shifting the image by two tooth widths, which leaves the position-only model right on under 9% of teeth, changes the detectors' top-1 by under 0.3 points. Masking everything beyond one box width costs Faster R-CNN and RT-DETR-l about 26 points on correctly numbered teeth (YOLOv8x 9.6), but among detected joint failures 74 to 93% keep the shared wrong number.

Next to missing teeth. 44.7% of joint failures have a missing neighbor, against 8.5% of teeth all three detectors get right (ratio 5.3, CI 4.2 to 6.6). 67 of the 72 joint failures next to a gap (93%) take the missing tooth's number. Teeth that only the position-only model gets wrong have a missing neighbor 10.5% of the time: gaps are not where position fails in general, but where position and the detectors fail together.

At closed gaps. Fig. 1. Across 372 missing teeth with both neighbors present, the space left between the neighbors separates gaps with a joint failure from gaps without one (AUROC 0.78, CI 0.70 to 0.84; median S 0.00 against 0.32). 56% of these errors sit at gaps whose neighbors touch, against 20% of other gaps. Each detector's own errors track closure as well (AUROC 0.74 to 0.79), more than the position-only model's (0.68).

An open gap is not enough. Erasing a tooth makes a correctly numbered neighbor take its number in 2.0% (YOLOv8x), 4.4% (RT-DETR-l) and 5.7% (Faster R-CNN) of cases, against 0 to 0.2% for both controls. But all three detectors do it to the same neighbor in only 4 of 2,260 cases (0.18%).

Closing the gap reproduces the shared error. For the same erased teeth, we moved the distal neighbor sideways until it touched the tooth on the other side. Of 1,114 moved teeth that all three detectors numbered right on the

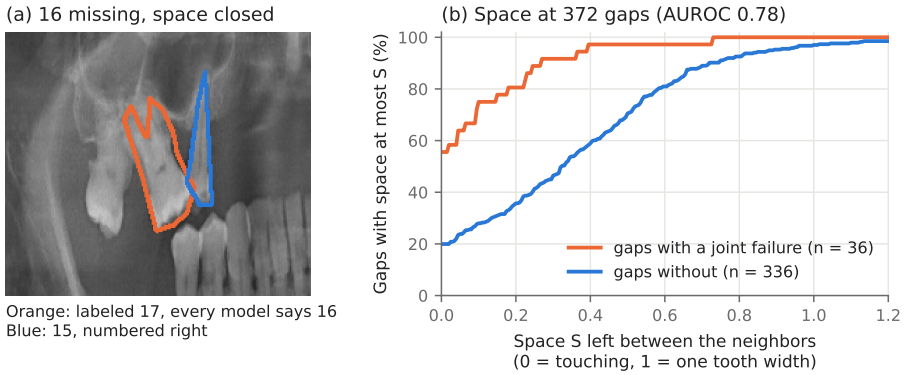


Fig. 1. (a) A closed gap: tooth 16 is missing and its neighbors' outlines touch. All three detectors and the position-only model number tooth 17 as 16. (b) Share of the 372 gaps with both neighbors present whose space is at most S, for gaps with and without a joint failure.

intact X-ray, all three gave the missing tooth's number to 0.18% with the space open, 2.9% when moved halfway and 20.8% when moved all the way (difference from open 20.7 points, CI 18.2 to 23.1); each detector alone did so 38 to 42% of the time. Models trained on two further splits give the same result (22.5% and 21.8% at full closure, at most 0.1% open). Tipping the neighbor about its root apex until its crown touches, which is closer to how real teeth drift, gives 9.4% (CI 7.7 to 11.0; 9.3% on the seed 1 models), near the 11% per neighbor at real closed gaps; cutting the tooth out and pasting it back unmoved gives 0.0%, so paste edges do not cause the effect. A moved tooth keeps its own shape, so the detectors number it by the slot it occupies.

5 Discussion

The detectors read teeth: they do not number by position in the frame, and they beat a position-only model on every tooth. Yet their rare shared errors are the position-only model's errors, and our results show where they happen. When a tooth is missing and its neighbors have moved together, the arch shows no gap. Counting along it gives the tooth next to the closed space the missing tooth's number, and that tooth also sits roughly where the missing one would. Appearance, context and position then point the same way, and every model follows. An inpainted gap leaves a full-width space and does not create this agreement; closing it does.

Two practical points follow. A position-only baseline is cheap and should be reported with tooth-numbering results: on standardized panoramic X-rays it sets a high floor, and the margin above it is what a model contributes. And the errors that different models share are not found by confidence; a review rule

from outside the model works better: check the numbering next to every missing tooth, especially where the space has closed. Here such a rule catches 97% of the shared errors at gaps but flags many gaps without one (precision 13%), since most gaps are partly closed. The rule can run on a model's own output: flagging teeth next to a position the detector left empty, or sharing a number, marks 12 to 16% of teeth and catches 41 to 47% of the shared errors on every split, against 17 to 32% for the same number of least-confident teeth (our pre-set 1.5-fold bar was met on two of three splits). Training does not remove the error easily: YOLOv8x retrained with simulated missing and closed teeth numbered teeth next to real gaps no better than a retrained control (+0.1 points, CI −1.6 to 1.7).

Limitations. One dataset from one source, a single annotation per tooth and no demographic data. On 634 DENTEX X-rays from other clinics [8], position alone numbers 66.9% of teeth. Trained within DENTEX, all three detectors beat it by 22 to 25 points, the shared errors give position-only's answer 85.5% of the time (our pre-set bar was 85.7%), and they sit at gaps 3.5 times as often as correct teeth (bar 4.2; 3.9 once DENTEX is weighted to UFBA's mix of missing teeth). A second training run on the same folds gives the same picture: margins of 24 to 26 points, 85.5% and a ratio of 3.3, with 237 of the 376 shared-error teeth in either run common to both. The UFBA-trained models, run unchanged, keep the position share (87.5%) but lose margin under the domain shift, mostly from missed teeth (YOLOv8x) or numbering one position too distal (Faster R-CNN). The direction of each finding holds on DENTEX; the gap link is weaker. Repeating the interventions on DENTEX X-rays with DENTEX-trained detectors (Supplement), moving the neighbor into the erased tooth's space gives the missing number 11.8% and 12.0% of the time on two training seeds (0.3% and 0.2% open), about 57% of the UFBA rate; the tip gives 4.8% and 5.5%, against our pre-set bar of 5%, met on one seed. The effect is present and smaller, and the individual teeth that fill differ between seeds (82 of 266 slide fills common to both). The interventions slide or tip a tooth within the image plane only (real teeth also rotate about their long axis), and masks on a 2D projection can touch when the teeth do not. Third-molar and longer gaps, which hold most missing positions, are outside the closure analysis. Two further splits, and retraining on the same split with a new seed, replicate the main numbers; about a third of the jointly failed teeth change with training randomness alone (overlap 0.63, against 0.54 to 0.57 across splits). About 10% of joint failures may be label errors.

6 Conclusion

On uncropped panoramic X-rays, position alone numbers 78.5% of teeth and detectors add 14 to 17 points. The few errors they share follow position and concentrate next to missing teeth whose neighbors have closed the space. On a second dataset, DENTEX, the same pattern holds with a weaker link to gaps and a smaller intervention effect. We release the per-tooth table and scorer so that new models can be measured against the baseline and on these shared failures.

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